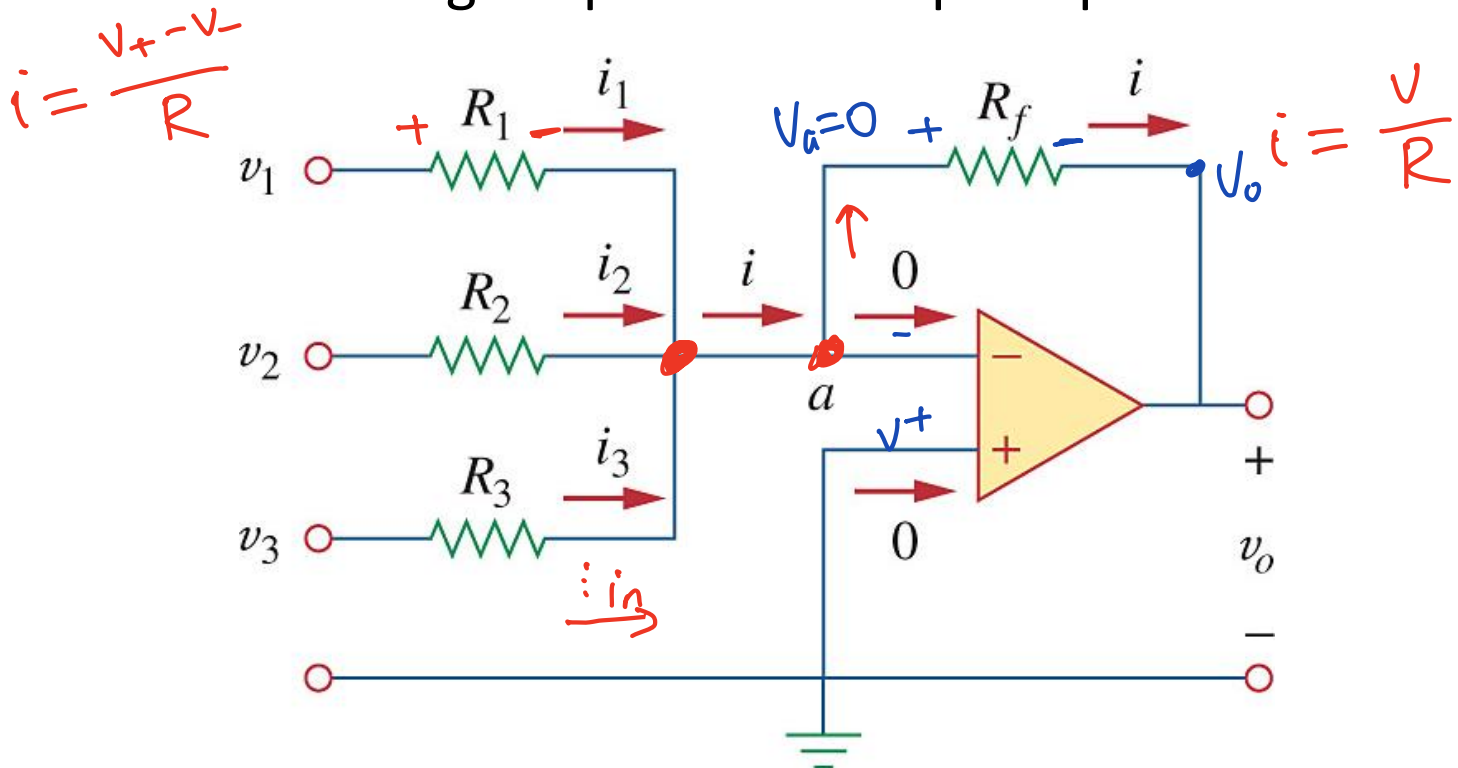


V_1

L#20: Summing and difference Amplifier

1. Configuration of summing amplifier

- Weighted sum of inputs; $V_o = k_1 V_1 + k_2 V_2 + k_3 V_3$
- Inverting Amps with multiple inputs.



(Give formula of V_o)

$$i = i_1 + i_2 + i_3$$

$$= \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = \frac{0 - V_o}{R_f}$$

$$V_o = k_1 V_1 + k_2 V_2 + \dots + k_n V_n = \sum_{i=1}^n k_i V_i$$

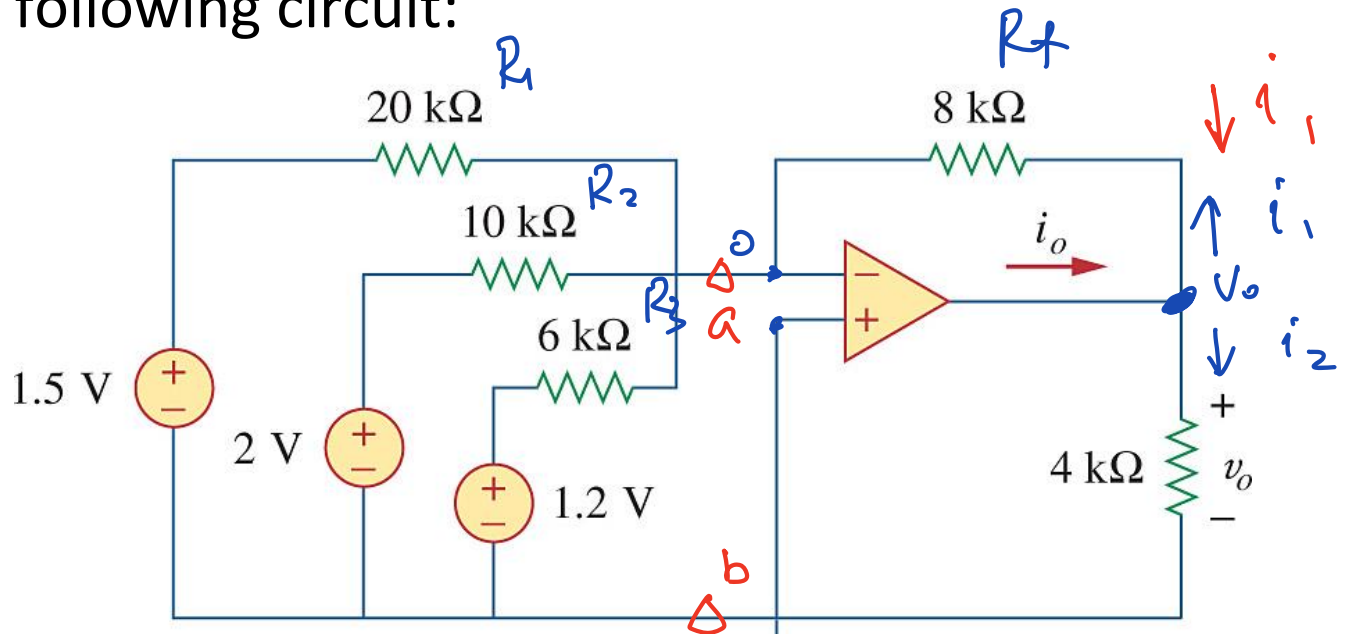
$$\frac{V_o}{R_f} = - \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$$

$$V_o = - \left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right)$$

$$\text{Let } R_f = R_1 = R_2 = \dots = R_n$$

$$V_o = - (V_1 + V_2 + \dots + V_n)$$

Example #1: Determine v_o and i_o for the following circuit:

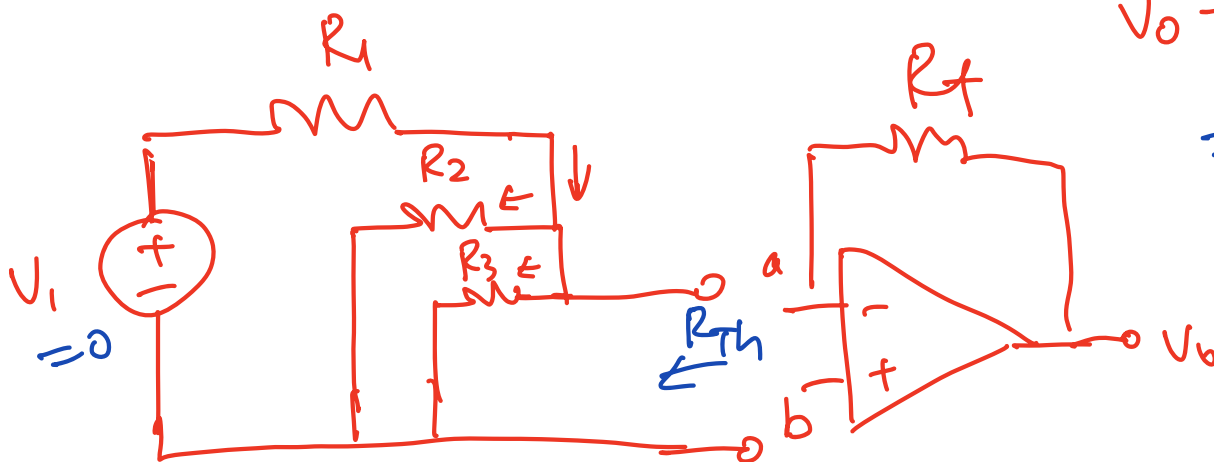


$$V_o = - \left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right)$$

$$= - \left(\frac{8}{20} \times 1.5 + \frac{8}{10} \times 2 + \frac{8}{6} \times 1.2 \right)$$

$$=$$

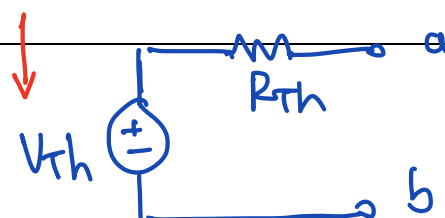
$$i_o = i_1 + i_2 = \frac{V_o - 0}{8k} + \frac{V_o}{4k}$$



$$\tilde{V}_o = - \frac{R_f}{R_1} \cdot V_1$$

$$= - \frac{R_f}{R_{Th}} \cdot V_{Th}$$

✓

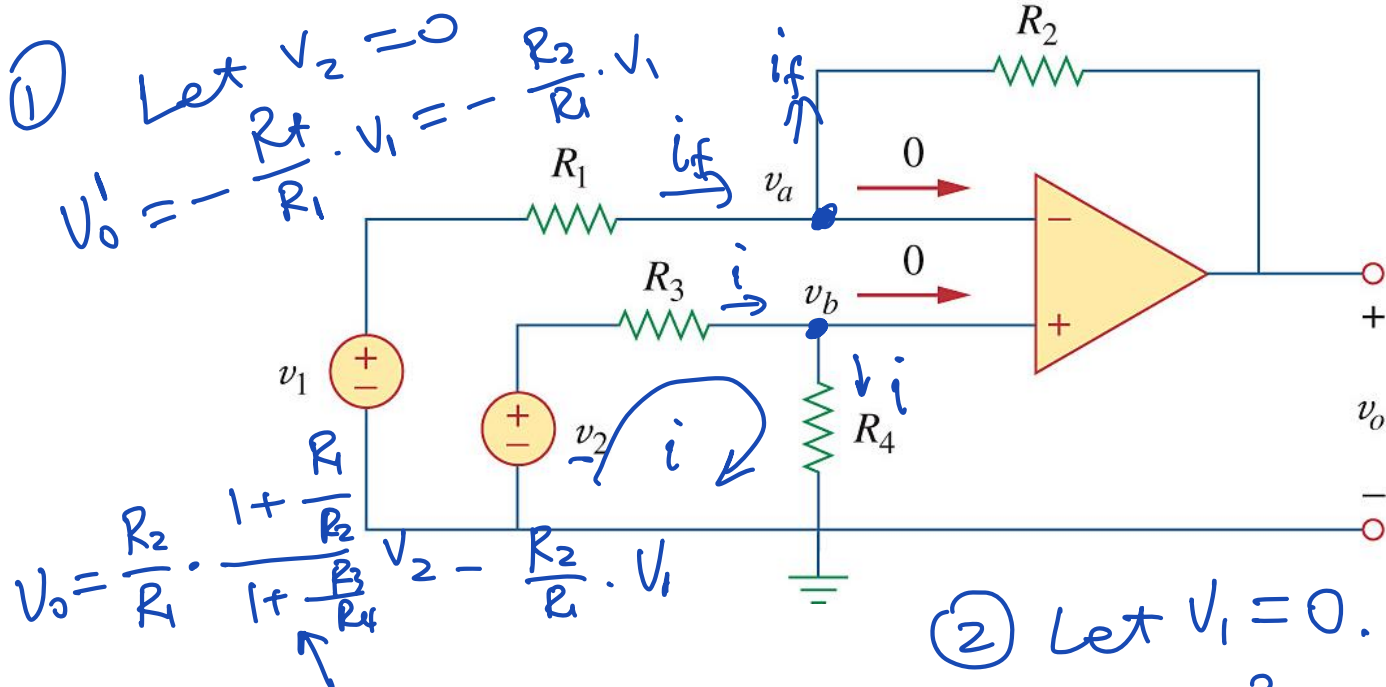


2. Configuration of difference amplifier

- Amplifies difference of two signals, but rejects signals in common;
- Combination of inverting and non-inverting Op Amps.

$$V_b = V_2 \cdot \frac{R_4}{R_3 + R_4}$$

$$V_a = V_b$$



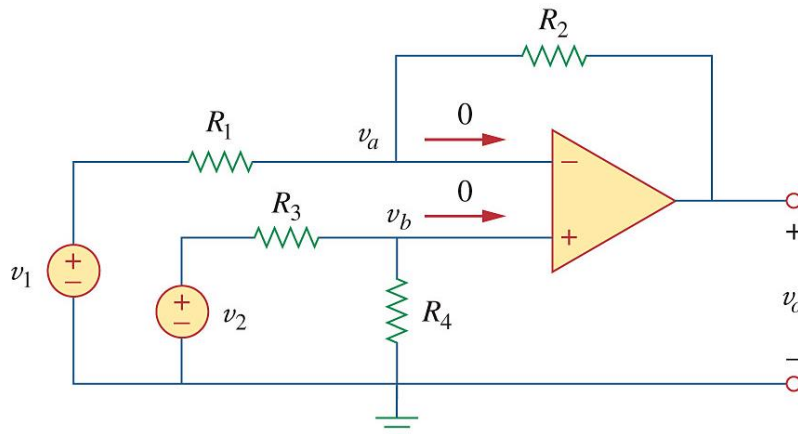
$$V_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} V_2 - \frac{R_2}{R_1} V_1$$

$$V_o^2 = V_i \left(1 + \frac{R_2}{R_1} \right), \quad V_i = V_b$$

$$= V_2 \cdot \frac{R_4}{R_3 + R_4} \cdot \frac{R_1 + R_2}{R_1} = V_2 \cdot \frac{R_2}{R_1} \cdot \left(1 + \frac{R_1}{R_2} \right) \cdot \frac{1}{1 + \frac{R_3}{R_4}}$$

$$V_o^T = \frac{R_2}{R_1} \cdot \frac{1 + \frac{R_1}{R_2}}{1 + \frac{R_3}{R_4}} \cdot V_2 - \frac{R_2}{R_1} V_1$$

Example #2: Design an Op Amp circuit with inputs v_1 and v_2 such that $v_o = -5v_1 + 3v_2$.



$$v_o = 3v_2 - 5v_1 \quad \Rightarrow \quad \frac{R_2}{R_1} = 5$$

$$= \frac{R_2}{R_1} \cdot \frac{1 + \frac{R_1}{R_2}}{1 + \frac{R_3}{R_4}} v_2 - \frac{R_2}{R_1} v_1$$

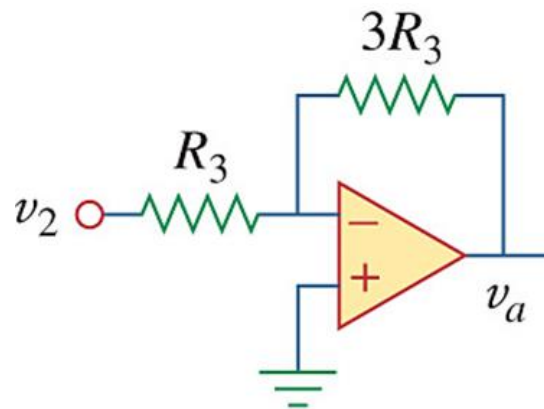
$$\begin{cases} R_1 = 1 \text{ k}\Omega \\ R_2 = 5 \text{ k}\Omega \end{cases}$$

$$= 5 \cdot \frac{1 + \frac{1}{5}}{1 + \frac{R_3}{R_4}} v_2 - 5v_1$$

$$\frac{6}{1 + \frac{R_3}{R_4}} = 3 \quad \Rightarrow \quad 1 + \frac{R_3}{R_4} = 2$$

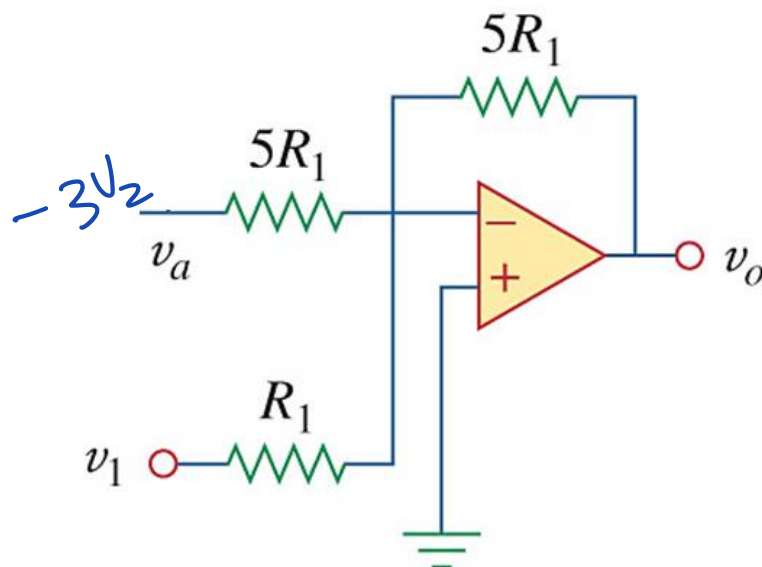
$$\Rightarrow R_3 = R_4 = 1 \text{ k}\Omega = 5 \text{ k}\Omega.$$

Example #2 (cont.): Design an Op Amp circuit with inputs v_1 and v_2 such that $v_o = -5v_1 + 3v_2$.



$$v_a = -\frac{3R_3}{R_3} \cdot v_2$$

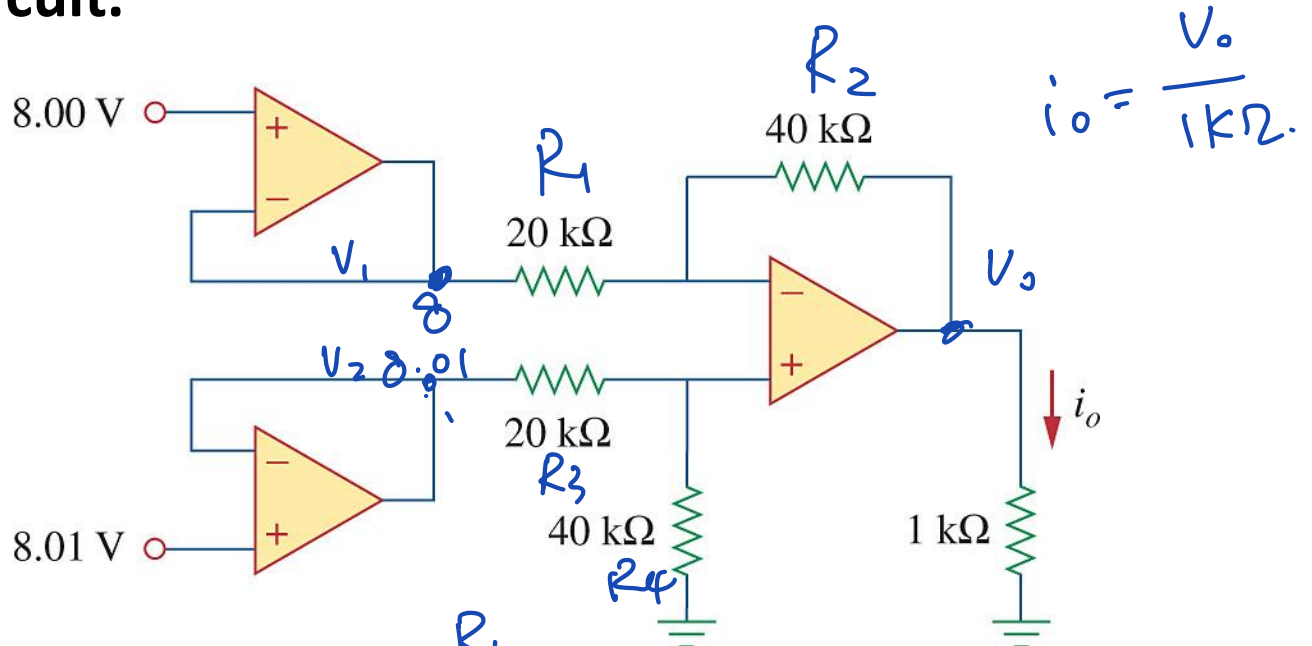
$$= -3v_2$$



$$v_o = -\left(\frac{5R_1}{R_1} \cdot (-3v_2) + \frac{5R_1}{5R_1} \cdot v_1\right)$$

$$= 3v_2 - 5v_1$$

Example #3: Determine i_o for the following circuit.



$$\begin{aligned}
 V_o &= \frac{R_2}{R_1} \cdot \frac{1 + \frac{R_1}{R_2}}{1 + \frac{R_3}{R_4}} \cdot 8.01 - \frac{R_2}{R_1} \cdot 8 \\
 &= 2 \cdot \frac{1 + 0.5}{1 + 0.5} \cdot 8.01 - 2 \times 8 \\
 &= 16.02 - 16 = 0.02 \text{ V.}
 \end{aligned}$$